

Effect of a mediterranean-style diet on endothelial dysfunction and markers of vascular inflammation in the metabolic syndrome: a randomized trial.

CONTEXT: The metabolic syndrome has been identified as a target for dietary therapies to reduce risk of cardiovascular disease; however, the role of diet in the etiology of the metabolic syndrome is poorly understood. OBJECTIVE: To assess the effect of a Mediterranean-style diet on endothelial function and vascular inflammatory markers in patients with the metabolic syndrome. DESIGN, SETTING, AND PATIENTS: Randomized, single-blind trial conducted from June 2001 to January 2004 at a university hospital in Italy among 180 patients (99 men and 81 women) with the metabolic syndrome, as defined by the Adult Treatment Panel III. INTERVENTIONS: Patients in the intervention group (n = 90) were instructed to follow a Mediterranean-style diet and received detailed advice about how to increase daily consumption of whole grains, fruits, vegetables, nuts, and olive oil; patients in the control group (n = 90) followed a prudent diet (carbohydrates, 50%-60%; proteins, 15%-20%; total fat, <30%). MAIN OUTCOME MEASURES: Nutrient intake; endothelial function score as a measure of blood pressure and platelet aggregation response to l-arginine; lipid and glucose parameters; insulin sensitivity; and circulating levels of high-sensitivity C-reactive protein (hs-CRP) and interleukins 6 (IL-6), 7 (IL-7), and 18 (IL-18). RESULTS: After 2 years, patients following the Mediterranean-style diet consumed more foods rich in monounsaturated fat, polyunsaturated fat, and fiber and had a lower ratio of omega-6 to omega-3 fatty acids. Total fruit, vegetable, and nuts intake (274 g/d), whole grain intake (103 g/d), and olive oil consumption (8 g/d) were also significantly higher in the intervention group ($P < .001$). The level of physical activity increased in both groups by approximately 60%, without difference between groups ($P = .22$). Mean (SD) body weight decreased more in patients in the intervention group (-4.0 [1.1] kg) than in those in the control group (-1.2 [0.6] kg) ($P < .001$). Compared with patients consuming the control diet, patients consuming the intervention diet had significantly reduced serum concentrations of hs-CRP ($P = .01$), IL-6 ($P = .04$), IL-7 ($P = 0.4$), and IL-18 ($P = 0.3$),

as well as decreased insulin resistance ($P < .001$). Endothelial function score improved in the intervention group (mean [SD] change, +1.9 [0.6]; $P < .001$) but remained stable in the control group (+0.2 [0.2]; $P = .33$). At 2 years of follow-up, 40 patients in the intervention group still had features of the metabolic syndrome, compared with 78 patients in the control group ($P < .001$). CONCLUSION: A Mediterranean-style diet might be effective in reducing the prevalence of the metabolic syndrome and its associated cardiovascular risk.